Calibration Admission Requires Mechanical Qualification: Four Complete Native Simulation Trials with Retained Failures

Research development draft [1]

Abstract

Physical-time scheduling is meaningful only when the preserved skill can produce the required effective events. Four frozen Isaac Sim/PhysX FR3 calibration trials complete all commands and pass independent raw consistency checks. Two lighter-payload trials complete manipulation; two heavier-payload trials lose transport grasp and have no stable-release endpoint. All four remain in the calibration denominator. Consequently no interval fit or held-out timing test is admitted. These retained negative observations expose a mechanical qualification gap rather than a demonstrated scheduling or learning improvement.

1. Protocol and complete result

The preserved Isaac Sim 5.1/PhysX A-r7 scene contains 56 rigid bodies and nine articulations. The FR3 executes the original bottle skill with physical mass/inertia factors 0.8 or 1.2 and a position-target FIFO of zero or 12 ticks at 240 Hz. Gravity compensation and state guards remain immediate. A prospective 118-file execution release, four-case calibration roster and original mechanical thresholds are fixed before trials. Every case is a fresh, uninterrupted native epoch; no task-period object or joint state write occurs.

Table 1. All four calibration cases, with missing endpoints retained.

Mass	FIFO	Samples	Mechanic	Release
0.8	0	13,427	TRUE	Observed
0.8	12	13,439	TRUE	Observed
1.2	0	13,427	FALSE	Missing
1.2	12	13,439	FALSE	Missing

The trials contain 53,732 task samples and 36 actual RGB snapshots. Resource allocation is bounded by 0.40953 GPU hours, including worker startup. All contact response intervals are approximately 171–175 ms. Light-case stable-release responses are 88–92 ms; heavy-case endpoints are absent. Calibration uses the original unscheduled skill, whose timing is early; command completion is not timing success. Four attempted calibration cases cannot be replaced by the two mechanically eligible cases. The interval fit is absent, and zero of the planned 16 timing tests has run.

2. Independent failure localization

Raw clocks, queues, controls, contacts, states, events and existing scores are independently recomputed. Both heavy cases acquire and lift the bottle, then lose relative-pose/contact continuity and tilt acceptance during transport. Only the zero-FIFO heavy case later collides with another bottle; collision is not the common failure cause. The paired robot TCP discrepancy is below 0.04 mm, while the bottle drift is large. This localizes grasp loss but does not establish a causal actuator or friction diagnosis.

For light cases, 13 strict-hold negative samples occur between commanded opening and composite stable-release onset, with none before opening. The unchanged diagnostic therefore cannot be interpreted as premature grasp loss. Main mechanics, event gaps and strict hold retain their prospective separate definitions.

An initial auditor incorrectly demanded bitwise equality between requested and installed inertia. Its rejection is retained. The new auditor applies the originally frozen installation tolerance (10^{-6} relative, 10^{-10} absolute for inertia); all state, queue, score and event checks remain exact. This audit-contract repair does not remove the two physical transport failures.

3. Core precedents and next hypothesis

MATTERIX already provides waiting, temporal process semantics, concurrent agents, workflow verification and HSM skills (Darvish et al., 2026). Its heating example distinguishes power-off from subsequent reaction completion. Auto-Engineering likewise provides engineering harnesses, BC/DAgger and deployment-artifact reuse. These are precedents rather than our new principles. A full paper must

[1]Negative-result component supplement; no full learning-system claim. Correspondence to: Author details pending.

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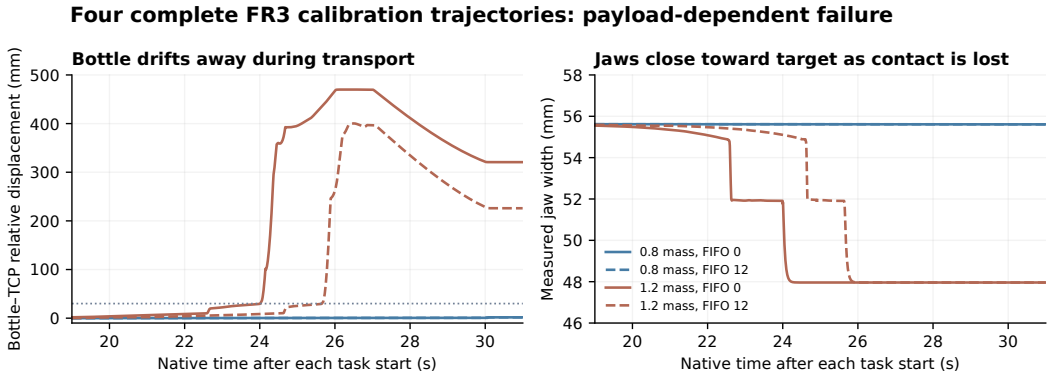


Figure 1. Actual native state observations from all four trials. Local task time aligns separate epochs; no cross-epoch continuity is asserted.

demonstrate independently calibrated tight contracts, valid recovery under an original external event, and shared actor absorption against strong HSM/scheduling and matched relabeling. The earlier twin-rotor counterfactual-reference component is not that external-event recovery test.

A separate candidate qualification is proposed: preserve the original scene/drive restoration, install and measure one hand-stiffness change before the task, and keep mass, targets, friction and mechanical criteria fixed. A subsequent separate one-case qualification passes these mechanics but lacks before-change damping/effort arrays and state-token preimages. It does not prove a causal root cause or provide a complete independent three-snapshot installation audit. A new source with complete logging must pass all four fresh calibration cases before timing tests. New success never rewrites the retained old failures. No new policy weights, chemical endpoints or real robot trials are credited here.

References

Darvish, K., et al. MATTERIX: toward a digital twin for robotics-assisted chemistry laboratory automation. *Nature Computational Science*, 6:67–82, 2026. https://doi.org/10.1038/s43588-025-00924-4.